INITIAL FYP PROPOSAL / 28 SEPTEMBER 2026

AI-Powered Multimodal Content & Assessment Generation

For Educational Applications

Proposed scope, user value and evaluation plan

> **Project aim.** Develop and evaluate a web application that turns course materials and teaching requirements into reviewable explanations, quizzes and assignments, with audio and visual support for learning.

The project will address educators preparing learning materials and university students undertaking independent study. It will investigate whether AI assistance can reduce preparation effort while maintaining acceptable educational quality. All capabilities and studies below are proposed; expected benefits will require evaluation.

1. Proposed scope

Core area	Planned deliverable
Course evidence	Import course materials; retrieve relevant passages; preserve source references. Retrieval-augmented generation (RAG) will supply selected course evidence to the generation model.
Content and assessment	Generate explanations, quizzes and assignments with answers and explanations. Accept learning objectives, prior knowledge, language, question type and requested numbers per difficulty level.
Multimodal transformation	Produce at least two additional output modalities: playable audio narration and usable instructional visuals. Each will be checked against the reviewed text.
Teacher and student use	Provide teacher editing, approval and export, and student access to approved materials for reading and self-practice. Basic access control will separate drafts from approved content.

2. Scope boundaries

The initial question types will be multiple choice and short answer. The application will support separate course workspaces; evaluation will focus on selected undergraduate computing courses rather than claiming validity across all disciplines. Tailoring will use stated needs and prior knowledge. Long-term learner modelling, automatic grading, secure examinations and full classroom administration will remain outside the core scope.

Scope basis: CEG 311 FYP project brief, AY 2026/2027, p. 27: *AI-Powered Multimodal Content & Assessment Generation for Educational Applications.* The user scenarios, boundaries and evaluation plan in this document are proposals for supervisor discussion.

Value for teachers and students

The following needs are starting hypotheses. Early interviews with teachers and students will check their importance, identify current practices and refine the requirements before implementation.

3. Teacher needs and expected value

Need to investigate	Proposed support	Expected value to test
Preparing materials from dispersed notes and readings	Draft an explanation or assessment from selected course sources and objectives.	Less repetitive work in locating, organising and drafting material.
Preparing exercises for different levels of prior knowledge	Specify learner background, question types and intended cognitive demand.	Easier preparation of varied practice with explicit teaching requirements.
Checking AI-generated questions and answers	Present reference answers, explanations and cited passages together; allow revision and approval.	Lower checking and correction effort at an acceptable quality standard.

4. Student needs and expected value

Relevant practice. Students will be able to study approved explanations and exercises linked to their course topics. An optional self-study workspace will allow requests based on their own materials; generated drafts will remain clearly identified as unapproved.

Explanations and self-checking. Answers will be hidden until requested, with explanations and source links available afterwards. This will support independent review without implying automatic assessment of the student.

Alternative presentation. Audio and instructional visuals will offer additional ways to access the same content. Their usefulness will be evaluated for specific tasks, without assuming that a preferred format necessarily improves learning.

5. Illustrative use case

> A teacher selects lecture notes on database transactions and requests six practice questions for students who know the basic definitions: two foundational, two application and two integrative questions, each with an answer and supporting evidence.

The proposed application would retrieve relevant passages, draft candidate questions and run automated checks. The teacher would inspect, edit and approve the material before making it available to students. Students could attempt the exercises, reveal explanations and revisit the supporting passages. Reviewed explanations could also be presented as narration or an instructional diagram.

Success in this scenario: the teacher obtains usable, course-aligned material with an acceptable amount of effort, and students can use it for meaningful practice. Generating six questions alone would not establish either outcome.

What AI would change

6. Comparison with a non-AI workflow

Without generative AI, teachers can already use documents, search tools, question banks and learning platforms to create high-quality materials. The proposed change is the ability to draft and revise new material in response to a particular combination of course evidence, learning objectives and learner background.

Activity	Without generative AI	With the proposed application
Create material	A person selects existing items or writes new explanations, questions and answers.	AI proposes new drafts from the supplied requirements and course evidence.
Adjust a task	A person changes the wording, scenario and reasoning requirements.	AI proposes revisions or variants for human inspection.
Check quality	A person checks the content against the source material.	Automated checks assist review; a teacher decides whether instructional material is acceptable.
Manage and distribute	Conventional software stores, versions and distributes materials.	The same software functions are connected to generation and review; they are not unique AI capabilities.

The expected advantage is a lower cost of preparing suitable variants and explanations. This is a hypothesis: if reviewing and correcting a generated draft takes longer than preparing it manually, an efficiency benefit would not be established.

7. Added value beyond ordinary chat-based AI

A general-purpose chat model can also draft questions, use supplied documents and revise answers. This project will investigate the value of a dedicated workflow: persistent course organisation, explicit output requirements, traceable evidence, automated checking, revision history and controlled approval. These mechanisms are intended to make repeated teaching tasks easier to manage and evaluate.

RAG will provide relevant evidence, but it will not guarantee factual accuracy. A valid source link may still fail to support a claim. Likewise, a requested difficulty label or a model judgement will not establish how difficult a question is for actual students.

8. Proposed research questions

RQ1 - Preparation effort and quality. Can AI assistance reduce the human effort needed to produce acceptable, course-aligned materials?

RQ2 - Value of the dedicated workflow. Does the proposed application improve evidence checking, revision or task completion compared with ordinary chat-based AI?

RQ3 - Student use and presentation. How useful and clear are the materials for independent practice, and do audio and visual formats preserve the instructional meaning?

The intended contribution will be an evaluated system design, a reproducible comparison and an analysis of failures and trade-offs. A new foundation model is not proposed.

Evaluation and planned delivery

9. Fair comparison

A pilot will compare three conditions: **A**, manual preparation using course materials and conventional tools; **B**, ordinary chat-based AI with access to the same materials; and **C**, the proposed application. Equivalent tasks will use the same objectives, output requirements and quality criteria. Task order will be varied to reduce practice effects.

Where practical, both AI conditions will use the same generation model. Additional checking calls will be included in the application's cost. Initial setup will be reported separately from repeated use. A separate comparison with and without retrieved evidence may examine the specific contribution of RAG.

Evaluation area	Planned measures
Material quality	Subject-expert review of anonymised outputs: correctness, course alignment, answerability, ambiguity, source support and intended cognitive demand. Difficulty targets will be hidden during initial rating.
Teacher effort and cost	Human preparation, checking and correction time; total elapsed time; revisions; usable completion rate; failed attempts and available API costs. Missing cost data will be reported as unknown.
Student experience	Task completion, clarity of explanations, ease of finding evidence and perceived usefulness. Claims about learning gains or actual difficulty would require a separate study with student responses.
Multimodal fidelity	Keep the underlying content constant and inspect narration and visuals for omissions, changed meaning, incorrect terminology and unsupported details; gather task-specific usability feedback.

Teachers and students will help define what counts as usable material before the main comparison. Participant numbers, recruitment and consent procedures will be agreed with the supervisor. A small pilot will establish feasibility; it will not be presented as conclusive evidence of effectiveness.

10. Planned stages and deliverables

Requirements and design. Confirm user needs, authorised course materials, acceptance criteria and the evaluation protocol. Using existing models through application programming interfaces (APIs) is proposed, subject to supervisor agreement.

CA1. Complete the literature survey and system design, and demonstrate an initial functional prototype with core text generation and RAG.

CA2 and final delivery. Integrate content and assessment generation with audio and visuals in a functional web application. Deliver the evaluated prototype, code, technical documentation, project report and reproducible evaluation records.

11. Items for supervisor agreement

Confirm the main teaching scenarios and pilot courses; the depth expected for audio and visual outputs; the acceptable model/API approach; and access to subject reviewers and student participants. These decisions will determine the detailed schedule and evaluation scale.